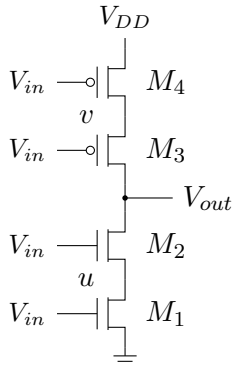
Introduction to VLSI Design, 2015

Homework 1: complete theory and CMOS solution

The original PDF, rather than OCR, determines the expressions. Q4(a) has an outer complement over the full expression and a second complement over $(x + y)$.

1. Which inverter transistors cannot saturate?

M_1 is the grounded NMOS, M_2 the NMOS near the output; M_4 is the PMOS adjacent to V_{DD} and M_3 the PMOS near the output. Let u be the NMOS internal node and v the PMOS internal node. All gates use V_{in}. Neglect body effect as specified.



If M_1 saturates, $u \geq V_{in} - V_{tn}$. Then $V_{GS,2} = V_{in} - u \leq V_{tn}$ and M_2 cannot carry nonzero current. Hence a conducting M_1 must be linear. Similarly, M_4 saturation implies $v \leq V_{in} + |V_{tp}|$, giving $V_{SG,3} = v - V_{in} \leq |V_{tp}|$ and switching off M_3. Thus

$$\boxed{M_1 \text{ and } M_4 \text{ cannot saturate while their series paths conduct.}}$$

M_2 and M_3 can saturate. This statement concerns nonzero conduction, not a cutoff device classified using inequalities alone.

2. Series-channel equivalence: complete derivation

Let the bottom NMOS have length L_2, the top NMOS length L_1, common width W, common threshold V_t, and gate voltage V_G. Set the external source to 0 and drain to V_D; the internal node is u. Define $k = \mu_n C_{ox} W$ and $a = V_G - V_t$. When both devices are linear,

$$IL_2/k = au - u^2/2,$$
$$IL_1/k = (a - u)(V_D - u) - (V_D - u)^2/2.$$

Adding cancels the internal node:

$$\boxed{I = \frac{k}{L_1 + L_2}\left(aV_D - \frac{V_D^2}{2}\right).}$$

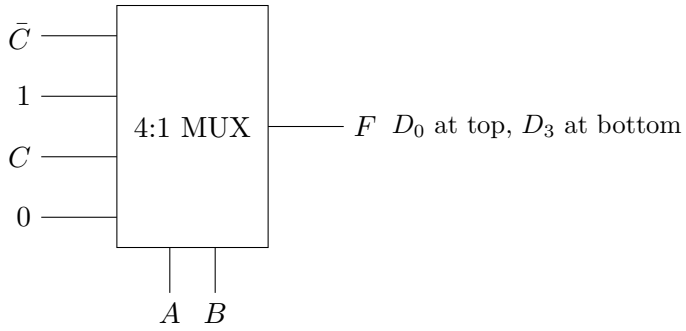
This is exactly the long-channel linear current of a single transistor with $W/(L_1 + L_2)$. If the upper transistor saturates, replace its equation by $IL_1/k = (a - u)^2/2$; adding the lower equation gives $I = ka^2/[2(L_1 + L_2)]$, the corresponding saturation current. For $L_1 = L_2 = L$, the equivalent device has width W and length $2L$. Common thresholds, no body effect, and the ideal square-law model are essential. The equivalence does not preserve internal diffusion capacitance or physical transient delay.

3. One 4-to-1 MUX implementation

The required function is $F = \bar{A}(B + \bar{C}) + A\bar{B}C$. Choose A, B as the select bits, with A the most significant bit. The Shannon cofactors are

A	B	Data input
0	0	$D_0 = \bar{C}$
0	1	$D_1 = 1$
1	0	$D_2 = C$
1	1	$D_3 = 0$

Thus $F = \text{MUX4}(A, B; \bar{C}, 1, C, 0)$. If $\bar{C}$ is unavailable, generate it with a CMOS inverter (2 extra MOSFETs).



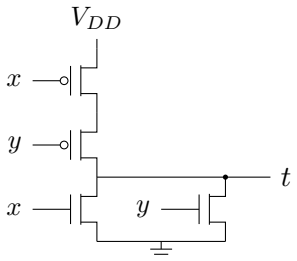
A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

4(a). Double-bar static CMOS function

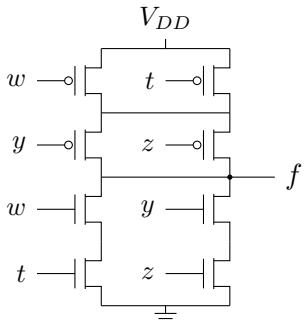
The visually confirmed expression is

$$\boxed{f = \overline{w\,\overline{(x+y)} + yz}.}$$

Generate $t = \overline{x+y}$ with a NOR gate (4 MOSFETs). Then a static AOI22 gate forms $f = \overline{wt+yz}$ using 8 more MOSFETs. No complemented external literal is required. The NOR PDN is $n_x \parallel n_y$ and its PUN is p_x in series with p_y. The AOI22 PDN has (n_w, n_t) and (n_y, n_z) series branches in parallel; its PUN is $(p_w \parallel p_t)$ in series with $(p_y \parallel p_z)$.



NOR: 4 transistors



AOI22: 8 transistors

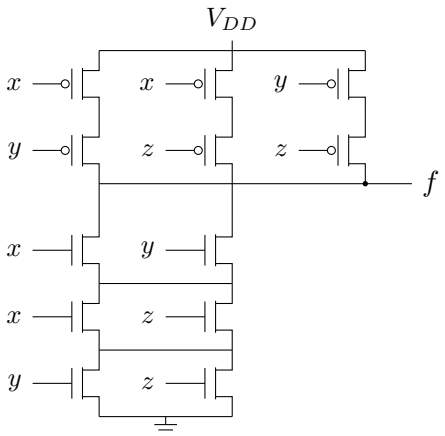
The intermediate t is the signal joining the two diagrams. PMOS body terminals connect to V_{DD}, NMOS bodies to ground. Total device count is 12. For all 16 true-input combinations, each CMOS stage has exactly one conducting pull-up or pull-down network.

4(b). Anti-majority static CMOS function

For $f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z}$, use

$$\bar{f} = (x + y)(x + z)(y + z) = xy + xz + yz.$$

The PDN is three series stages of two parallel NMOS devices. The PUN is its dual: three parallel branches of two series PMOS devices. Gate labels use only true inputs; repeated labels are separate devices. This direct construction has 12 transistors.



Source review and checks

Both supplied handwritten JPGs were inspected. Their M_1/M_4 answer and series-current elimination agree with Q1/Q2. Their 4-to-1 MUX table agrees with Q3. The drawn NOR intermediate plus CMOS network matches Q4(a); the rewritten algebra in the scan is less clear than the original PDF's two overbars, so the circuit was independently checked against the original expression. The Q4(b) diagram is retained as a valid direct dual network.

Run `python verify_logic.py`. All 8 MUX, 16 Q4(a), and 8 Q4(b) combinations pass in `truth_tables.csv`. These are exhaustive Boolean conduction checks, not electrical simulation.